

Problem Set 2: The Schrödinger Equation

Introduction

The purpose of this problem set is to *introduce* you to a few ideas related to the material we covered on the Schrödinger equation and quantum mechanics. I suggest that you begin with a review of your class notes, employing the lecture-note summary as a review guide. **Please do this now.**

There are four problems in this problem set.

- Problem 1 introduces you to a numerical solution of the Schrödinger equation using the *method of finite differences*. Its aim is to use the classical problem of a particle in an escape-proof “box” to illustrate some of the numerical issues that crop up when trying to find a quantum entity’s allowed energies.
- Problem 2 gives you a chance to see how easily the method of finite differences can be used to solve the radial equation of hydrogen, a task which is much more difficult to do analytically.
- Problem 3 will demonstrate to you the beautiful and powerful way in which quantum mechanics uses *appropriate mathematical operations* to extract *accessible physical information* from wave functions.
- Problem 4 gives you a chance to *apply* concepts from the earlier problems, together with what you’ve already learned in Problem Set 1. You’ll see how simple energy information for an electron in a molecule, which this problem set has shown you how to get, at least in principle, together with the rate-equation model we studied in Problem Set 1, can be used to predict a current-voltage feature known as *negative differential resistance*. It’s worth noting that this feature has actually been measured in the laboratory by workers in the field of *molecular electronics*.

As you do the work in this problem set, you may want to note the key results or concepts covered by each problem; that’ll tell you what you should understand for exams.

Problem 1 (Numerical Solution for the Particle in a Box)

- (a) Consider the problem of a particle in an escape-proof “box,” as we discussed it in the lectures. **Write** a Python code to find the eigenvectors and eigenvalues of the corresponding matrix equation $[\hat{\mathcal{H}}]\{\phi\} = E\{\phi\}$ that we developed. Use lattice points from $n = 1$ to $n = N$, with $N = 100$ and a lattice constant $a = 1 \text{ \AA}$. Note that the box width will then be $L = (N + 1)a$, since there are *implied* lattice points at $\ell = 0$ and $\ell = N + 1$, where we apply the boundary conditions.¹

Plot the position-probability densities from the numerical wave functions $\{\phi\}$ for eigenvalue numbers 1 and 50.

Plot the 100 eigenvalues that result from the numerical solution along with those from the analytical result

$$E = \frac{\hbar^2 \pi^2 n^2}{2mL^2} \quad (1)$$

where n is a positive integer.

A sample code and sample plots are provided for you at the end of this problem set.

Submit your code and plots.

- (b) Recall that the analytical solution can be written as

$$\phi(x) = A \sin\left(\frac{n\pi}{L}x\right) \quad (2)$$

where A is a constant, and where the “box” extends from $x = 0$ to $x = L$.

- (i) Given that *normalization* requires

$$\int_0^L |\phi(x)|^2 dx = 1 \quad (3)$$

find the value of A required to properly normalize the analytical wave function (2).

Hint: It may help for you to know that

$$\int \sin^2(\alpha x) dx = \frac{1}{2}x - \frac{1}{4\alpha} \sin(2\alpha x) \quad (4)$$

Answer: $A = \sqrt{2/L}$.

- (ii) For sufficiently small values of the lattice constant a , it is reasonable to assume that the ℓ th component of the numerical eigenvector $\{\phi\}$ will follow a functional form that is consistent with (2), and will be given, for example, by

$$\phi_\ell = B \sin\left(\frac{n\pi}{L}x_\ell\right) \quad (5)$$

¹Look again at your notes if this is unclear.

where B is a normalization constant and $x_\ell = a\ell$ is the ℓ th lattice point. If the numerical eigenvectors are normalized according to the equation

$$\sum_{\ell=1}^N |\phi_\ell|^2 = 1 \quad (6)$$

then **find** the expected value of B in the limit that $a \rightarrow 0$. *Hint:* Start by multiplying both sides of (6) by a , ask yourself what happens to the sum on the left-hand side as $a \rightarrow 0$, and then exploit the work you did in part (i). *Answer:* $B = \sqrt{2a/L} = \sqrt{a} \times A$.

The **main point** to note here is that even if the numerical eigenvectors have the proper functional form (5), they will differ from the analytical result (2) by a factor of $\sqrt{a}$, because they are normalized according to (6), which isn't perfectly consistent with (3).²

- (c) From the graph of energy eigenvalues in part (a), you can see that the numerical results begin to deviate significantly from the analytical results for $n \gtrsim 50$. Let's investigate this in a few different ways.

- (i) Recall the ℓ th line from the matrix equation $[\hat{\mathcal{H}}]\{\phi\} = E\{\phi\}$ reads

$$-t_0\phi_{\ell-1} + (2t_0 + U_\ell)\phi_\ell - t_0\phi_{\ell+1} = E\phi_\ell \quad (7)$$

where $t_0 \equiv \hbar^2/2ma^2$ and $U_\ell \equiv 0$ in the present problem. If ϕ_ℓ has the form (5), then **find** an expression for E that satisfies (7). *Hint:* Work on the left side of (7), making use of the trigonometric identities

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta \quad (8)$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta \quad (9)$$

and then compare the result to the right side, hence deducing E . *Answer:* $E = 2t_0\{1 - \cos[(n\pi a/L)]\}$.

- (ii) The answer to part (i) tells us how to expect the numerical eigenvalues to behave with the eigenvalue number n , assuming (5) is true.

Plot the answer; modify the code for part (a) by adding the following Python code to the appropriate location:

```
plt.plot(n, 2*t0*(1-np.cos(n*np.pi*a/L)), 'o', markerfacecolor='none')
```

Print the plot, and by referring to it, **comment** on how the answer to part (i) compares with the numerical results you obtained in part (a).

- (iii) Recall $\cos \theta \approx 1 - \theta^2/2$ for small θ ; then, apply this approximation to the answer of part (i) to **find** the expected numerical eigenvalues when $n\pi a/L$ is small. In our case, since a and L are fixed, we're basically finding the expected numerical eigenvalues when n is sufficiently small. **State** whether the result you get is consistent with what you see in the plot of part (ii). *Answer:* $E = (\hbar^2\pi^2n^2)/(2mL^2)$.

²As a result, do you see that a plot of *numerical* probability density, like that of $|\phi_\ell|^2$ in (5), should be strictly dimensionless, whereas a plot of *analytical* probability density, like that of $|\phi(x)|^2$ in (2), should have the units of $1/m$? Although this is the case, I will take the liberty of labeling the axes of all graphs with the units corresponding to **analytical** probability density, even when I'm plotting **numerical** results [like in the graphs for part (a)].

- (iv) The reason for the errors in the numerical eigenvalues at large n is because the true wave functions (2) start to get very oscillatory, and the lattice spacing a is too large to provide an adequate sampling of the oscillations; in other words, the sampled values specified by (5) might be correct [to within a factor of $\sqrt{a}$, as examined in part (b)], but they are too far apart. The situation can be improved by reducing the lattice spacing a .

Rerun the code of part (a), **but use** a value of a that is 10 times smaller, and a number of points N that is 10 times larger [so that the well size $L = (N + 1)a$ stays *approximately* the same], and **print** the plots for the probability densities and the energy eigenvalues. Make sure you **leave** the axis limit commands in line 73 intact,³ so that you can more easily compare the result to the corresponding plot in part (a).

Briefly comment on the behavior of the numerical versus analytical eigenvalues as a function of n , contrasting what you see with what you saw in part (a). Also **comment** on any perceived changes in the numerical probability densities for $n = 1$ and $n = 50$; focus on the *shapes*, and don't worry about the values, which will change since a has been changed, as examined in part (b).

- (d) In this part, we'll explore the impact of periodic boundary conditions, changing the numerical solutions from those of a particle in a "box" to those of a particle in a "ring," a problem that we also examined in class. We won't try to do a full comparison of numerical and analytical results, because the bookkeeping becomes unwieldy.

- (i) Begin by **deleting** line 73, and lines 75–87 in the sample code of part (a).

Then, **insert lines to modify** the Hamiltonian matrix $[\hat{\mathcal{H}}]$ to correspond to *periodic boundary conditions*.

Also **modify the code** to plot the probability densities for eigenvalue numbers 4 and 5, rather than 1 and 50. (recall that Python is indexed starting at zero, so the n -th eigenvector is obtained by indexing $\Psi[:, n-1]$).

Make sure you **use the same values** of a and N as in part (a).

By the way, I don't consider any of this to be "coding." The requested changes should be trivial, provided you've absorbed what we discussed in class.

Submit your modified code, and the plot for the probability densities, as well as the plot for the numerical eigenvalues.

Sample plots are provided for you at the end of this problem set, so that you can verify your work.

³Obviously, all the line numbers I mention refer to the sample code I provided in part (a); you should perform the suggested actions on the appropriate line numbers in *your* code.

- (ii) **State** the numerical eigenvalues (energies) for eigenvalue numbers 4 and 5, and **indicate** the proper units. You can retrieve and print the numbers by adding the following Python code to the end of your script:

```
print(E_numerical[4-1])  
print(E_numerical[5-1])
```

State whether or not the energy is *degenerate* for these eigenvalue numbers, and **justify** your answer.

- (iii) Using commands similar to those suggested in part (ii), **obtain** the energies for the numerical eigenvalue numbers 1 through 5, and **sketch** an *energy-level diagram* which clearly shows the number of available states at each energy. You can assume degeneracy for eigenvalues 2 and 3, and use your answer to part (ii) to decide on degeneracy for eigenvalues 4 and 5.
- (iv) Recall that, in class, we found the analytical eigenvalues for the particle in a “ring” to be given by

$$E = \frac{2\hbar^2\pi^2n^2}{mL^2} \quad (10)$$

where n is the analytical eigenvalue number. If $n = 0$ corresponds to numerical eigenvalue number 1, $n = 1$ corresponds to numerical eigenvalue numbers 2 and 3, and $n = 2$ corresponds to numerical eigenvalue numbers 4 and 5, then **calculate** the analytical values and **state** how they compare with the numerical results you found in part (iii).

Problem 2 (Radial Equation for the Hydrogen Atom)

The solution of the Schrödinger equation for the hydrogen atom is a three-dimensional problem. However, by viewing the problem in spherical coordinates, it turns out that the total wave function can be written as a product of a so-called *radial* part (dependent on the variable r) and an angular part (dependent on the polar and azimuthal angles of spherical coordinates). In this problem, we will look for solutions to the radial part.

The radial part can be written in terms of a function $f(r)$, such that $|f(r)|^2 dr$ represents the probability of finding the electron in the volume between r and $r + dr$, where the nucleus is at the origin $r = 0$.

The function $f(r)$ is governed by the *radial Schrödinger equation*, which can be derived from the full three-dimensional Schrödinger equation, and which can be written as follows:

$$\left[\frac{-\hbar^2}{2m} \frac{d^2}{dr^2} - \frac{q^2}{4\pi\epsilon_0 r} + \frac{l_o(l_o + 1)\hbar^2}{2mr^2} \right] f(r) = E f(r) \quad (11)$$

where $\epsilon_0 = 8.854 \times 10^{-12}$ F/m is the permittivity of free space, l_o is called the *orbital or angular quantum number*, and the other symbols have their usual meanings.

- (a) Let (11) be rewritten in the form

$$\hat{\mathcal{H}} f(r) = E f(r) \quad (12)$$

with

$$\hat{\mathcal{H}} \equiv \left[\frac{-\hbar^2}{2m} \frac{d^2}{dr^2} - \frac{q^2}{4\pi\epsilon_0 r} + \frac{l_o(l_o + 1)\hbar^2}{2mr^2} \right] \quad (13)$$

If (12) and (13) are to be solved using a spatial lattice with r being discretized, then **state** the values of the elements to be found on the main diagonal, upper diagonal, and lower diagonal of the matrix $[\hat{\mathcal{H}}]$ that would represent $\hat{\mathcal{H}}$.

- (b) If the boundary conditions are such that $f(r) = 0$ at the endpoints of the discretized space, then **state** the values of the upper-right and lower-left, corner entries of $[\hat{\mathcal{H}}]$.
- (c) **Modify** your code from Problem 1(a) of this problem set to solve the radial equation (11), assuming $f(r) = 0$ at the endpoints of the discretized space. Use a lattice spacing $a = 0.1$ Å with 100 points, and let $l_o = 0$, and **plot** the probability densities $|f(r)|^2$ for the first two eigenvectors, which will correspond to the so-called “1s” and “2s” energy states. Sample plots are provided at the end of this problem set.

Hints: In the sample code for Problem 1(a), you should change the value of a in line 17, and (for clarity) the name of the position variable from x to r in line 18. You may also want to define a variable that carries the value of $\epsilon_0 = 8.854 \times 10^{-12}$ F/m, and appropriately rename a few of the other variables (*i.e.*, rename `phi_numerical_1` to `f_1` and so on), such that your code reads cleanly. Lines 29–32 should be modified according to your answer to part (a) above, and when doing so, *please be careful to enter the expression for $\cup$ in the proper units of eV*. You should modify lines 50–66 to find and plot $|f(r)|^2$ for the first two

eigenvectors. **Use two separate graphs**, with the axes scaled (using the `plt.xlim` and `plt.ylim` commands) to **match** those in the sample plots. Finally, delete lines 68–85.

Submit your code and the plots—but do part (f) below before printing your plots.

- (d) **State** the numerical value of the eigenvalue (or *eigenenergy*) for the 1s level, which you can obtain and print by appending the following line of Python code to the end of your script:

```
print(E_numerical[0])
```

Indicate the proper units.

- (e) The analytical expression for the 1s eigenfunction is

$$f(r) = \frac{2r}{a_0^{3/2}} e^{-r/a_0} \quad (14)$$

where $a_0 = 4\pi\epsilon_0\hbar^2/mq^2 = 0.0529 \text{ nm}$ is known as the *Bohr radius*.

By demanding that this eigenfunction satisfy the radial Schrödinger equation with $l_o = 0$, **analytically calculate** the corresponding eigenenergy, and **compare** the result to the numerical value from part (d). *Answer:* $E = -13.6 \text{ eV}$.

- (f) On your numerical plot of $|f(r)|^2$ for the 1s level, **superimpose** a plot⁴ of $a \times |f(r)|^2$, with $f(r)$ given by the analytical expression (14). You can do this as follows, assuming you plotted your numerical result in figure 1 in Python, and assuming you changed the variable `x` to `r`, as discussed in part (c). Append the following Python code to the figure 1 section of your code:

```
plt.plot(r, a*( 2*r / 0.0529e-9**(3/2) *
              np.exp(-r/0.0529e-9) )**2, 'o', markerfacecolor='none')
```

State how the analytical result behaves compared to the numerical result, and whether or not this is to be expected based on how the analytical eigenenergy you found in part (e) compares with the numerical result from part (d).

⁴The extra factor of a is needed to account for differences in normalization between the numerical and analytical results, as examined in Problem 1(b).

Problem 3 (Momentum for the Particle in a Box)

In this problem, we'll see how to obtain *momentum information* for the particle in an escape-proof "box." This will highlight some interesting and useful aspects of quantum mechanics.

- (a) Recall from your work in Problem 1 that the ground-state ($n = 1$) wave function can be written as

$$\phi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi}{L}x\right) \quad (15)$$

where the box extends from $x = 0$ to $x = L$.

Before doing anything, for this problem, let's define a *new* position axis x' , and let the box now lie between $x' = -L/2$ and $x' = +L/2$. In essence, we're making a change of variable,

$$x' = x - L/2 \quad (16)$$

You can see (16) works, since when $x = 0$ (start of box), we now have $x' = -L/2$, and when $x = L$ (end of box), we get $x' = L/2$.

Combine (15) and (16) to **find** the mathematical form of the ground-state wave function on the x' -axis.

The use of your answer and the x' -axis will prove more convenient for this problem than the use of the original form (15) and the x -axis. Of course, all *physical information* will remain unaffected; we're just executing a mathematical adjustment to ease the upcoming bookkeeping.

Answer: $\phi(x') = \sqrt{2/L} \cos(\pi x'/L)$.

- (b) Things are going to get a little strange here, so get yourself ready.

As we discussed in class, all physically accessible information that can be known about a quantum particle (entity) can be obtained by performing the *appropriate* mathematical operations on its wave function. In this case, we have in $\phi(x')$ the wave function that applies when the particle possesses its ground-state energy $E = \hbar^2\pi^2/2mL^2$, and by performing the appropriate mathematical operations on this wave function, we can hence get whatever (accessible) information we want about the particle when it has that energy. But just what are these "appropriate mathematical operations"? Let's take a look at an example where we want to know about the particle's *momentum*.

It turns out, i.e., I'm stating this without proof, that you can get momentum information by taking the *Fourier transform* of the wave function $\phi(x')$. That's right, the Fourier transform! What exactly does this mean? In electrical engineering, you've already encountered the Fourier transform as a means of expressing a time-domain signal in terms of its frequency-domain spectrum. In that case, the transform variables were time t and radian frequency ω . Now, and get ready for this, we'll define the transform variables as being position x' and

momentum⁵ p' . So, a wave function of position $\phi(x')$ will be transformed into a wave function of momentum, let's call it $\Phi(p')$. The latter function $\Phi(p')$ can then be interpreted as a *momentum-probability* distribution, just like $\phi(x')$ is a *position-probability* distribution; in other words, $|\Phi(p')|^2 dp'$ represents the probability of obtaining a momentum that is within dp' of the value p' in a momentum measurement, just like $|\phi(x')|^2 dx'$ represents the probability of finding the particle in a space dx' about x' in a position measurement.⁶ You better stop here and make sure you follow what I'm saying; try only to absorb the result for now, because I haven't explained the rationale.

So, let's go ahead and define the Fourier transform. First, let me recall the time-frequency transform equation,⁷ which you've seen before, for a signal $f(t)$ and its Fourier transform $F(\omega)$:

$$F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \quad (17)$$

Now, to emphasize that what we're about to do is take a Fourier transform, let me introduce the intermediate variable k' , and an intermediate function $A(k')$, and write

$$A(k') = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(x') e^{-ik'x'} dx' \quad (18)$$

Now look at (17) and (18). Do you see that just as (17) defines a transform between functions of time t and the variable ω (which has units of inverse time), that (18) defines an *analogous* transform between functions of position x' and the variable k' (which has units of inverse position)? In other words, $A(k')$ is the *Fourier transform* of $\phi(x')$. Can we attach any more meaning to $A(k')$? Yes, and the starting point is to consider the inverse transform equations:

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega \quad (19)$$

$$\phi(x') = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k') e^{ik'x'} dk' \quad (20)$$

The inverse transform relation (19) to which you are accustomed says that $F(\omega)$ carries information on the complex amplitudes of an infinite sum of complex exponentials $e^{i\omega t}$ that are used to reconstruct the signal $f(t)$, and that the sum is over all radian frequencies ω . Now, the inverse transform relation (20) says an analogous thing: $A(k')$ is an *amplitude function* that carries magnitude and phase information on an infinite sum of *plane-wave*, wave functions $e^{ik'x'}$ that are used to *reconstruct* the original wave function $\phi(x')$, and the

⁵Since I introduced a prime for the position variable, I'm also going to use a prime on the momentum variable, because it makes the equations look prettier; hence, we have x' and p' as our variables.

⁶In case you are wondering, for the present problem, the mentioned probabilities are independent of time, and can hence be found by starting with the time-*independent* part of the wave function $\phi(x')$, instead of the full wave function $\Psi(x', t)$.

⁷It's worth mentioning that I'll define the Fourier transform in a *symmetric* fashion, in the sense that *both* the forward and reverse transform equations will have multiplying prefactors of $1/\sqrt{2\pi}$. In some treatments, especially in electrical engineering, the prefactor is omitted from the forward transform (17) while the reverse transform (19) gets a prefactor of $1/2\pi$. This isn't really important, but you do have to know how the prefactors are arranged before using *somebody else's* tables of transform pairs.

sum is over all wave vectors k' . You might want to read those last two sentences again, slowly.

Now, to get the momentum information we want, I'll do a little bookkeeping. Let me first execute a change of variable, and define $p' \equiv \hbar k'$. This basically says that associated with every plane-wave, wave function appearing in the infinite sum (20) is a specific momentum $p' = \hbar k'$, a fact that will be justified in part (d) of this problem. Moreover, let me multiply the answer in (18) by $1/\sqrt{\hbar}$. These steps don't change the fact that whatever we finally obtain *still* originates from (18), and can hence justifiably be called a Fourier transform of $\phi(x')$. Performing the indicated operations, we end up with the so-called *wave function in momentum space*, as follows:

$$\Phi(p') \equiv \frac{1}{\sqrt{\hbar}} A\left(\frac{p'}{\hbar}\right) \quad (21)$$

Wow, that was a lot of material to absorb. So, let me tell you the recipe once more: start with a wave function $\phi(x')$ in the so-called *position space* or *position representation*; take its Fourier transform à la (18) to get an amplitude function $A(k')$; multiply the result by $1/\sqrt{\hbar}$ and execute a change of variable $p' = \hbar k'$ or $k' = p'/\hbar$ as indicated in (21). You'll end up with what's known as the wave function $\Phi(p')$ in momentum space or *momentum representation*. *Capisci?*⁸

- (i) The wave function for the ground state has the functional form given by the answer to part (a) *only* in the region $-L/2 \leq x' \leq L/2$. To mathematically indicate this, we should *really* write the function as follows:

$$\phi(x') = \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x'}{L}\right) \times \text{rect}\left(\frac{x'}{L}\right) \quad (22)$$

where

$$\text{rect}\left(\frac{x'}{L}\right) = \begin{cases} 0 & x' < -L/2 \\ 1 & -L/2 \leq x' \leq L/2 \\ 0 & x' > L/2 \end{cases} \quad (23)$$

Now consider the following Fourier-transform results involving the variables t and ω :

$$\mathcal{F}[\text{rect}(t/\tau)] = \frac{1}{\sqrt{2\pi}} \tau \text{sinc}\left(\frac{\omega\tau}{2\pi}\right) \quad (24)$$

$$\mathcal{F}[f(t) \cos(\omega_1 t)] = \frac{1}{2} [F(\omega + \omega_1) + F(\omega - \omega_1)] \quad (25)$$

where $\mathcal{F}$ denotes the transform operation, and where $\text{sinc } x \equiv (\sin \pi x)/\pi x$. Use these relations and your knowledge of Fourier transforms, but with x' and k' playing the roles of t and ω , respectively, to write the Fourier transform $A(k')$ of the wave function in (22) as the sum of two sinc functions centered at $\pm k_1 = \pm\pi/L$. *Answer:*

$$A(k') = \frac{1}{2} \sqrt{\frac{L}{\pi}} \left\{ \text{sinc}\left[\frac{L}{2\pi}(k' + k_1)\right] + \text{sinc}\left[\frac{L}{2\pi}(k' - k_1)\right] \right\}$$

⁸That's Italian for, "Do you understand?"

- (ii) **Execute** the prescription in (21) to obtain the momentum wave function $\Phi(p')$, again written as the sum of two sinc functions, this time centered at $\pm p_1 = \pm \hbar k_1 = \pm \hbar \pi / L = \pm \sqrt{2mE}$, where $E = \hbar^2 \pi^2 / 2mL^2$ is the ground-state energy. Hence, notice that the sinc functions are centered at the values of *classical* momenta we'd expect if the energy was E .⁹ *Answer:*

$$\Phi(p') = \frac{1}{2} \sqrt{\frac{L}{\pi \hbar}} \left\{ \text{sinc} \left[\frac{L}{2\pi \hbar} (p' + p_1) \right] + \text{sinc} \left[\frac{L}{2\pi \hbar} (p' - p_1) \right] \right\}$$

- (iii) **State** the units of $|\Phi(p')|^2$, and **state** and why the answer is to be expected.
- (iv) By starting with the answer to part (ii), **compute** the value of the integral

$$\int_{-\infty}^{\infty} |\Phi(p')|^2 dp'$$

and **state** why the result you get is to be expected. You'll need the following properties of the sinc function:

$$\int_{-\infty}^{\infty} \text{sinc}^2 y dy = 1 \quad (26)$$

$$\int_{-\infty}^{\infty} \text{sinc}(\kappa - y) \text{sinc}(y) dy = \text{sinc}(\kappa) \quad (27)$$

$$\text{sinc}(-y) = \text{sinc}(y) \quad (28)$$

Note: This part requires you to do some manipulation with integrals. Please be patient, and work through it step-by-step. I'm asking you to do it in order to drive home the point regarding the meaning of $|\Phi(p')|^2$.

- (v) **Download** the Python code needed to create plots of the two sinc-function components, call them $\Phi_1(p')$ and $\Phi_2(p')$, in the answer to part (ii), as well as their sum $\Phi(p')$, including the multiplying prefactor of $(1/2)\sqrt{L/\pi\hbar}$. Note that it is possible to create such plots in this case because the sinc-function components and $\Phi(p')$ are all real.¹⁰

The code also plots the probability density $|\Phi(p')|^2$ in a separate figure.

Note that the plots employ a *normalized momentum* axis p'/p_1 . **Warning:** You might be tempted to interpret the horizontal axis of the plots in terms of the position x' or the box width L , *but of course you cannot do this because the horizontal axis is (normalized) momentum p'/p_1 . The plots contain no information whatsoever related to position.* If you're confused by what I just said, back up and review the last few parts of this problem, and *think* about what is being plotted.

Print the plots.

Just to make sure everything worked fine, you can compare your plots to the samples provided at the end of the problem set.

⁹The values of classical momenta can be found by setting $mv^2/2 = E$, where v is velocity, and rearranging to obtain $mv = \pm \sqrt{2mE}$.

¹⁰In general, of course, $\Phi(p')$ could be complex.

(vi) On your plots, **mark** the locations of classical momenta $\pm\sqrt{2mE}$ with vertical lines.

If the particle is an electron with its free mass $m = 9.11 \times 10^{-31}$ kg, then **state** the magnitude of the electron's velocity at the points of classical momenta. You'll need to know that the plots were created with a value $L = 101$ Å, which was chosen to be consistent with Problem 1.

(vii) **Comment** on the validity of the following statement: "For a particle in the ground state of an escape-proof 'box,' the only possible momenta are the classical values $\pm\sqrt{2mE}$."

(c) By inspection of the plot of $|\Phi(p')|^2$ in part (b)(v), **state** the *average value* of momentum, *i.e.*, the expectation value of momentum. *Think about it.* Look at the plot and ask yourself, "If I had a large number of identical particles, each in its own box and each in the ground state, and I measured the momentum for each particle, what would be the *average* value of the measurements?" *Hint:* Remember that $|\Phi(p')|^2$ represents a probability distribution function, and hence the average value is defined as

$$\langle p' \rangle = \int_{-\infty}^{\infty} p' |\Phi(p')|^2 dp' \quad (29)$$

Then, *deduce* the result of the integration from the shape of your graph, without doing any calculations.

Now, you can verify your prediction. Rather than work on the integration in (29), in this case, it's easier to work with $\phi(x')$ and employ the use of a so-called *momentum operator*. The average value of momentum can be found as follows:

$$\langle p' \rangle = \int_{-\infty}^{\infty} \phi^*(x') \hat{p} \phi(x') dx' \quad (30)$$

where $\hat{p} \equiv -i\hbar d/dx'$ is the "momentum operator" and the superscript * denotes the complex conjugate.¹¹ It turns out that this is equivalent to (29). Use (30) along with (22) to **find** the average momentum, and **verify** that it agrees with your prediction from (29) and the plot of $|\Phi(p')|^2$. The following integral may prove useful:

$$\int \sin(\alpha x') \cos(\alpha x') dx' = \frac{1}{2\alpha} \sin^2(\alpha x') \quad (31)$$

(d) In quantum mechanics, every observable quantity has associated with it an *operator*. For a given wave function, *i.e.*, for the particle in a given state, the value of the observable is said to be *sharp* if the wave function satisfies the eigenvalue equation associated with the operator. For example, for the momentum operator, if we have a wave function $\theta(x')$, then the momentum is said to be "sharp" if

$$\hat{p} \theta(x') = p' \theta(x') \quad (32)$$

¹¹Obviously, you'll have to take this on faith. But, remember, all physical information in quantum mechanics is found by doing the appropriate mathematical operations on wave functions. So, I'm just *telling you* the required operation on $\phi(x')$.

for some number p' . When this happens, then the *only* associated value of momentum with that wave function, *i.e.*, the only value obtainable in a measurement when a particle is described by that wave function, is p' .

Suppose we have a plane-wave, wave function $\theta(x') = e^{ik'x'}$. **State** whether or not the momentum associated with this wave function is sharp and, if so, **state** its value p' . Your answer should tell you why the change of variable leading to (21) is justified.

So, did all that blow your mind? **You learned that momentum information can be found by taking an appropriately adjusted Fourier transform of the wave function $\phi(x')$** , illustrating one way that quantum mechanics *ruthlessly* deploys mathematical tools to extract physical information.¹²

¹²If you haven't already picked up on this point, quantum mechanics is ruled by the mathematics of *linear algebra*, involving such things as linear operators, linear differential equations, eigenvalues, and eigenfunctions. By contrast, Newtonian mechanics is often said to be ruled by the mathematics of *differential and integral calculus*.

Problem 4 (An Application: Negative Differential Resistance)

This problem will give you an opportunity to apply what you've learned in this problem set, along with what you learned in Problem Set 1. We'll examine a phenomenon called *negative differential resistance*. I will keep the hints to a minimum, so you'll have trouble unless you've really absorbed all the material up to this point.

- (a) To begin, consider an electron in a diatomic molecule. Such a molecule can often be (crudely) modeled with what's known as a *simple-harmonic-oscillator* or *SHO* potential-energy profile. In this case, the potential energy of the electron as a function of position x is given by $U(x) = (1/2)m\omega_0^2x^2$, where ω_0 is known as the *SHO frequency* of the molecule, and the origin $x = 0$ lies halfway between the two atoms in the molecule. The eigenenergies for the first two levels of such a potential have been experimentally measured, and are known to be given by $(1/2)\hbar\omega_0$ and $(3/2)\hbar\omega_0$, respectively.

Consider a diatomic molecule for which $\hbar\omega_0 = 0.2879$ eV. **Download** the code required to solve for the eigenvectors and eigenenergies of such a molecule. *Before running the code*, you must **insert** an appropriate value of the lattice constant a . Given that the position grid has been arranged to lie between $x = -31$ Å and $x = +31$ Å, **choose and state** an appropriate value of a that will yield reliable plots for the position-probability densities for the first two eigenstates, and **justify** your answer.

Submit the plots generated by the code for the position-probability densities.

- (b) The position-probability densities that you obtained in part (a) apply for an electron in the respective eigenstates of an *isolated* molecule. This information may be useful, but to deal with electron flow, of course, what we really need is the energy information. From part (a), you now have both experimental and numerical results for the energies.

The next step is to utilize the energy information to model the electron flow. As a nanoelectronics engineer, to get a quick idea of the current-voltage behavior, you decide to use the simple rate-equation model we developed in the first part of the course, and that you studied in Problem Set 1.

A physicist tells you that, for the molecule in question, he has reason to believe the following:

- Only the first two energy levels are important for electron flow. Upon coupling to the contacts, the levels *smear*, and can be represented by Lorentzian functions, as expected; however, in addition, the second level *lowers* its center energy by 0.1 eV.
- The coupling coefficient of the source contact is energy dependent, with $\gamma_1(E) = 0.005$ eV for $E > 0$, and $\gamma_1(E) = 0$ for $E \leq 0$. The drain coupling coefficient is $\gamma_2(E) = 0.005$ eV for all energies.

The physicist also gives you all the other information you need for the rate-equation model: $\alpha_G = \alpha_D = 0.5$, $U_0 = 0.025$ eV, $k_B T = 0.025$ eV for room-temperature operation, and $\mu = 0$.

In order to predict the current in the two molecular levels, you get a “code jockey” to modify one of your codes from Problem Set 1. He does this for you, and writes a code to solve the usual electrostatic equation together with the following transport equations:

$$D_{1,2}(E) = D_{\mathcal{E}_{1,2}}(E) = \frac{[\gamma_1(E) + \gamma_2(E)]/2\pi}{(E - \mathcal{E}_{1,2})^2 + \{[\gamma_1(E) + \gamma_2(E)]/2\}^2} \quad (33)$$

$$D(E) = D_1(E) + D_2(E) \quad (34)$$

$$N = \int_{-\infty}^{\infty} \frac{\gamma_1(E)f_1(E) + \gamma_2(E)f_2(E)}{\gamma_1(E) + \gamma_2(E)} D(E - U) dE \quad (35)$$

$$I = \frac{q}{\hbar} \int_{-\infty}^{\infty} \frac{\gamma_1(E)\gamma_2(E)}{\gamma_1(E) + \gamma_2(E)} [f_1(E) - f_2(E)] D(E - U) dE \quad (36)$$

where $\mathcal{E}_1$ refers to energy level 1, and $\mathcal{E}_2$ refers to energy level 2.

Your job as a nanoelectronics engineer is to use the physicist’s information and the code jockey’s code to *explain* the current-voltage behavior for $V_D \geq 0$.

- (i) **Download** the code from the website; there are two files, a main program and a subplot program, and you’ll need both. In the main program, **enter** the *appropriate* energies for the first two levels, based on the physicist’s information.

Run the code, and verify that you get the proper current-voltage curve, a sample of which is given at the end of this problem set; **print** and **submit** the plot.

- (ii) The code also produces a number of other plots. With the aid of these, and noting that for an energy level to carry current, its *shifted* density of states, $D_{\mathcal{E}_1}(E - U)$ or $D_{\mathcal{E}_2}(E - U)$, must lie where there is a “difference in agenda” between the contacts and where the overall coupling is nonzero, **state** the following:

- * which levels (1 or 2 or both or none) carry the current when $0 \lesssim V_D \lesssim 0.25$ V;
- * why the current abruptly drops to zero for 0.25 V $\lesssim V_D \lesssim 0.4$ V;
- * which levels (1 or 2 or both or none) carry the current when 0.4 V $\lesssim V_D \lesssim 0.65$ V;
- * why the current abruptly drops to zero for 0.65 V $\lesssim V_D \lesssim 0.8$ V;

You don’t necessarily need to print out all the plots, nor do you need to write long explanations; just write enough to show that you understand what is happening.

- (iii) A circuit designer wants to investigate the use of this molecule for the purposes of “negative differential resistance,” or “NDR,” which is said to occur whenever dI/dV is negative. **State** the approximate values of V_D over which NDR is present.

On the very last page of this problem set, I’ve appended a few diagrams from work that investigates NDR in real molecules through theory and experiment. *Don’t worry about the details*, and just note the following: diagram (a) shows a so-called “TEMPO” molecule attaching itself to silicon, which acts as one contact in the experiment, while the other contact is an STM (scanning tunneling microscope) tip; diagram (b) shows an energy-band diagram, much like the ones we’ve been using, and the modeled current-voltage behavior; and diagram (c) shows the experimentally measured current-voltage behavior. So, what do you think of *that*?

Conclusions

There were a lot of things we did in this problem set. Problems 1 and 2 gave you some practice in solving the Schrödinger equation using the *method of finite differences*, and explored associated numerical issues. Problem 3 *exposed* you to the power and beauty of quantum mechanics, of which the Schrödinger equation is merely one aspect. And Problem 4 gave you a chance to apply the concepts from this problem set together with those of Problem Set 1, *i.e.*, to combine energy information from a solution of the Schrödinger equation together with equations for electron flow, in order to predict and explain distinctive features of the current-voltage behavior of a nanodevice.

We won't need any heavy-duty quantum mechanics in our course, but it's important to note that a knowledge of the subject for engineering purposes comes down to two things: (1) how to solve the Schrödinger equation in a variety of situations for a quantum (nano) entity's wave function and allowed energies; and (2) how to use mathematical tools (like the Fourier transform) and/or mathematical pictures (like an energy-level diagram) to *appropriately* extract desired physical information from the wave functions and allowed energies. This problem set has given you a *taste* of both things, so you don't need to be afraid of the quantum physicist who tries to razzle-dazzle you—whatever he or she does, you can take heart that they're not really doing anything more than what you learned here.¹³

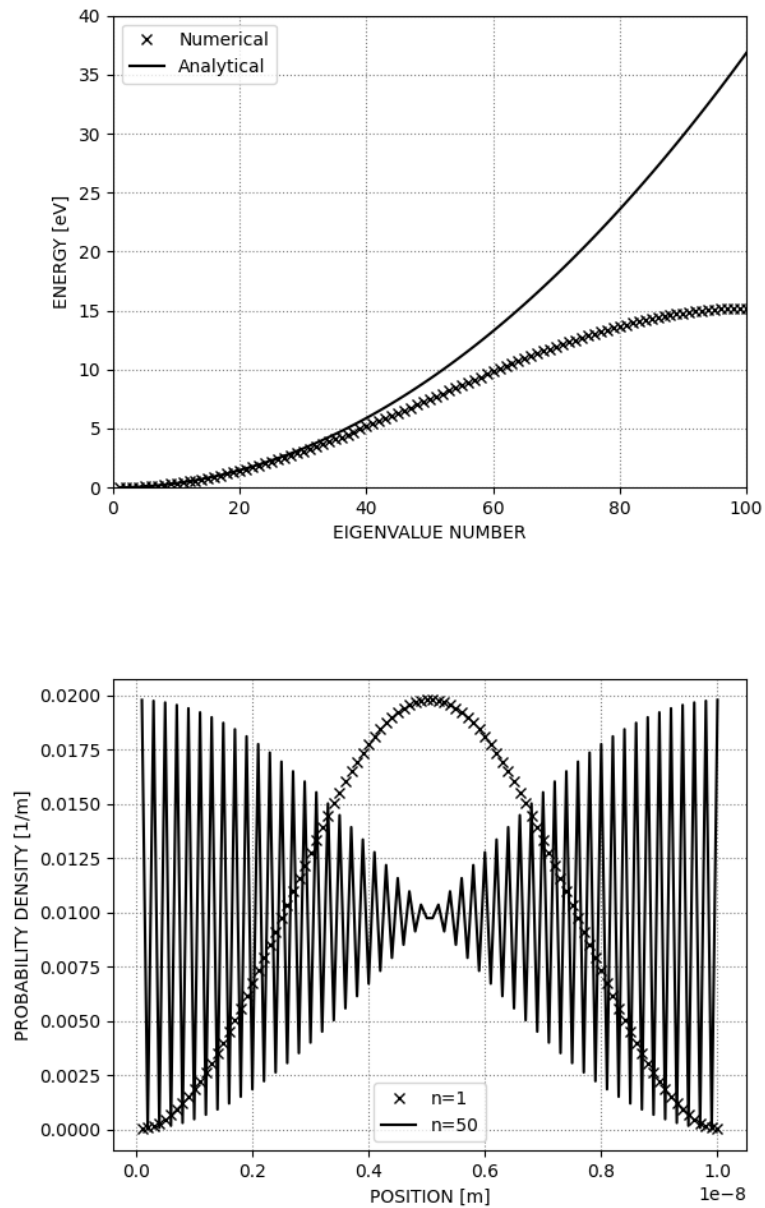
¹³On the other hand, if *you* wanted to impress a quantum physicist, say at a cocktail party, you could now drop a bit of jargon into the conversation. You might say something like, “You know, I think I might just have the eigenfunction to satisfy your eigenvalue equation.” Or even, “So, what operations do I need to perform to extract desired physical information from your wave function?” Think of it as a fringe benefit of taking this course.

Sample Code for Problem 1(a)

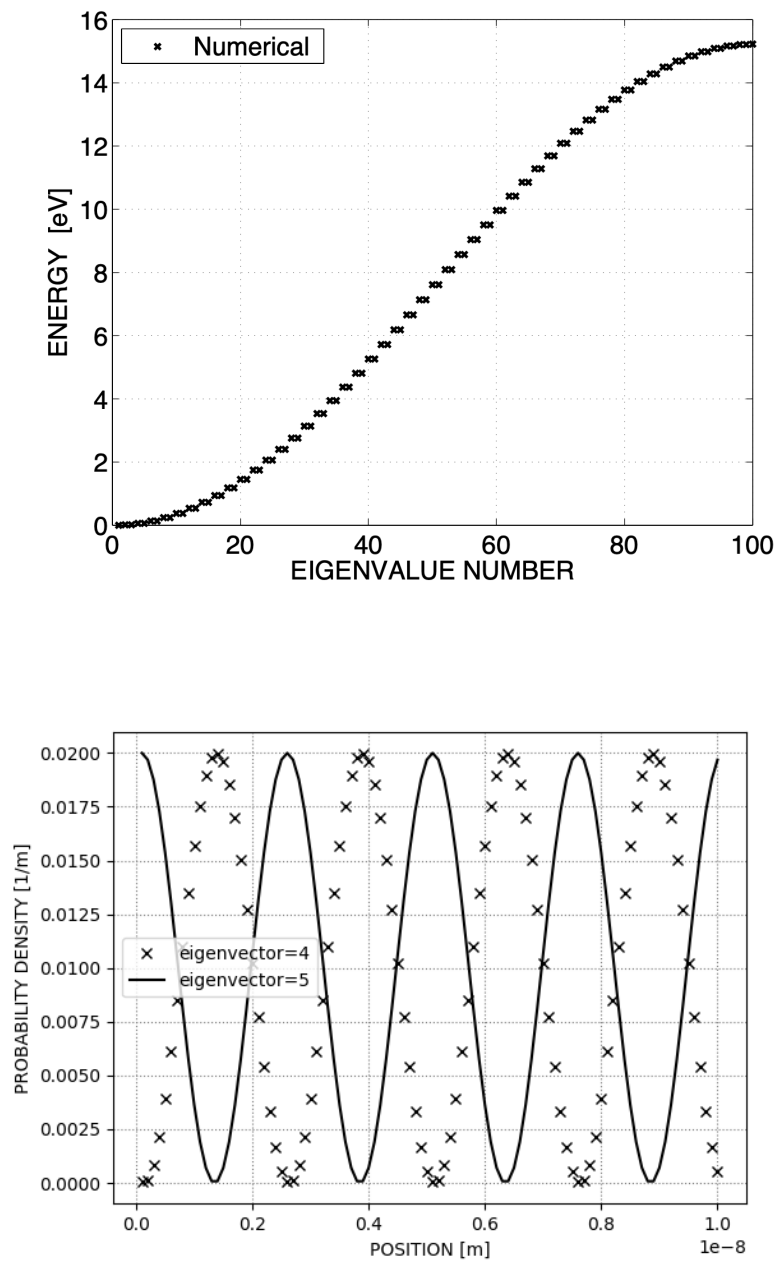
```
1  import numpy as np; import matplotlib.pyplot as plt
2
3  # Physical constants in MKS units, including free-electron mass m
4
5  hbar = 1.054e-34
6  q = 1.602e-19
7  m = 9.110e-31
8
9  # Lattice data: total number N of lattice points; an integer
10 # grid from n=1 to n=N; lattice constant "a"; a discretized position
11 # grid x; Hamiltonian parameter t0, as introduced in class, but now
12 # divided by q so that it has units of eV; total length of the lattice,
13 # L=a*(N+1)
14
15 N = 100
16 n = np.linspace(1,N,N)
17 a = 1e-10
18 x = a*np.linspace(1,N,N)
19 t0 = (hbar**2)/(2*m*a**2)/q
20 L = a*(N+1)
21
22 # Set up the Hamiltonian matrix, in pieces, by constructing square
23 # matrices containing information for the main
24 # diagonal, lower diagonal, and upper diagonal, and then adding them
25 # together; you may need to look up the diag command using help
26 # to see how this works; U is the potential inside the well, set
27 # to zero at each lattice point in this problem
28
29 U = 0*x
30 main_D = np.diag((2*t0*np.ones(N)+U),0)
31 lower_D = np.diag(-t0*np.ones(N-1),-1)
32 upper_D = np.diag(-t0*np.ones(N-1),+1)
33
34 H = main_D + lower_D + upper_D
35
36 # Find the eigenvectors and eigenvalues of H; V will contain the
37 # eigenvectors in its columns, and D will contain the corresponding
38 # eigenvalues on its main diagonal
39
40 D, V = np.linalg.eig(H)
41
42 # The resulting eigenvalues (and the list of corresponding eigenvectors) are not
43 # sorted by default, so we will sort them from min to max eigenvalue
44
45 index = D.argsort()
46 D = D[index]
47 V = V[:,index]
48 E_numerical = D
49
```

```
50 # Find the eigenvectors corresponding to eigenvalues 1 and 50;
51 # we can get these from the matrix V
52
53 phi_numerical_1 = V[:,0]
54 phi_numerical_50 = V[:,49]
55
56 # Find the position-probability densities
57
58 P_numerical_1 = phi_numerical_1 * np.conj(phi_numerical_1)
59 P_numerical_50 = phi_numerical_50 * np.conj(phi_numerical_50)
60
61 # Plot the probability densities
62
63 plt.figure(1); plt.plot(x,P_numerical_1,'kx',x,P_numerical_50,'k-')
64 plt.grid(color='grey', linestyle=':')
65 plt.xlabel('POSITION [m]'); plt.ylabel('PROBABILITY DENSITY [1/m]')
66 plt.legend(['n=1','n=50'])
67
68 # Plot the numerical eigenvalues
69
70 plt.figure(2); plt.plot(n,E_numerical,'kx',label='Numerical')
71 plt.grid(color='grey', linestyle=':')
72 plt.xlabel('EIGENVALUE NUMBER'); plt.ylabel('ENERGY [eV]')
73 plt.xlim(0,100); plt.ylim(0,40)
74
75 # Find the first N analytical eigenvalues, for comparison with the
76 # numerical results; note the prefactor of (1/q) to get the answers
77 # in eV
78
79 E_analytical = (1/q)*(hbar**2*np.pi**2)*(n*n) / (2*m*L**2)
80
81 # Superimpose the analytical eigenvalues onto figure 2
82 #
83
84 plt.plot(n,E_analytical,'k-')
85 plt.legend(['Numerical','Analytical'])
86
87 plt.show()
```

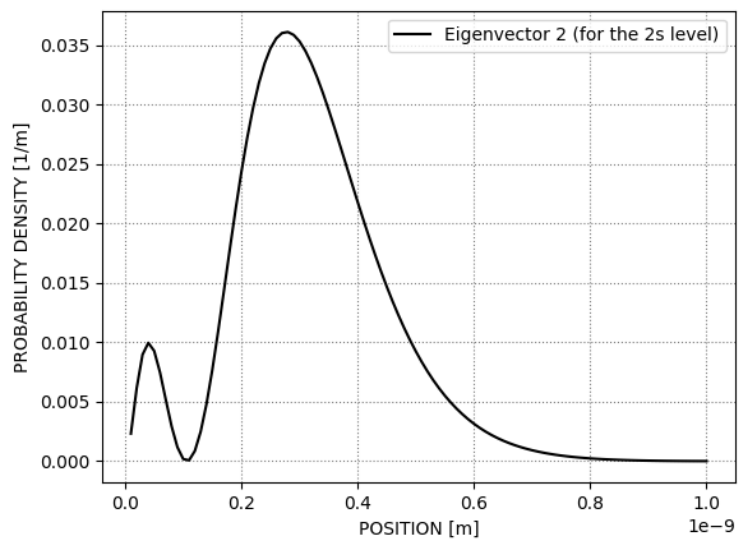
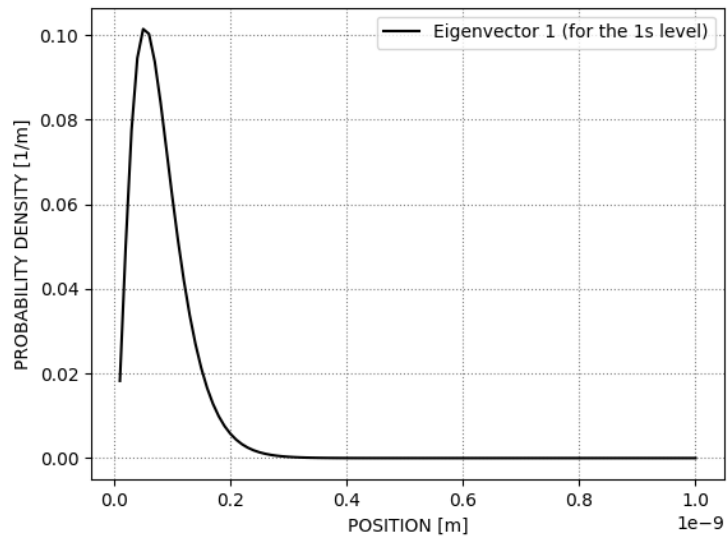
Sample Plots for Problem 1(a)



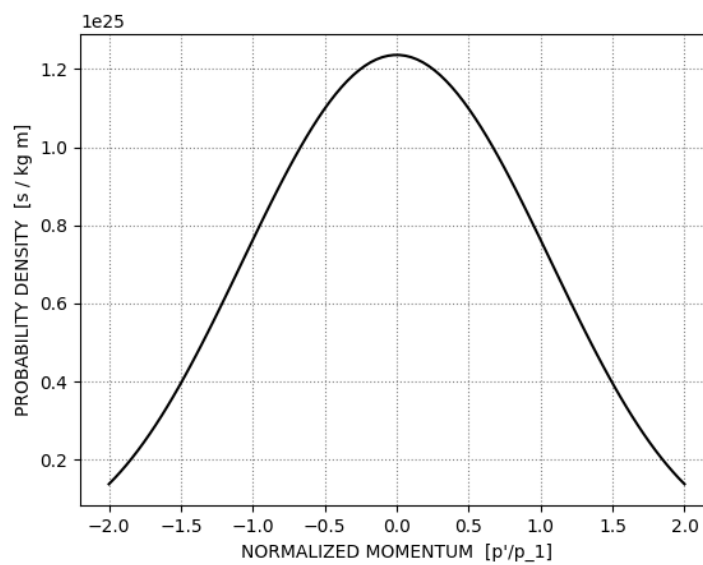
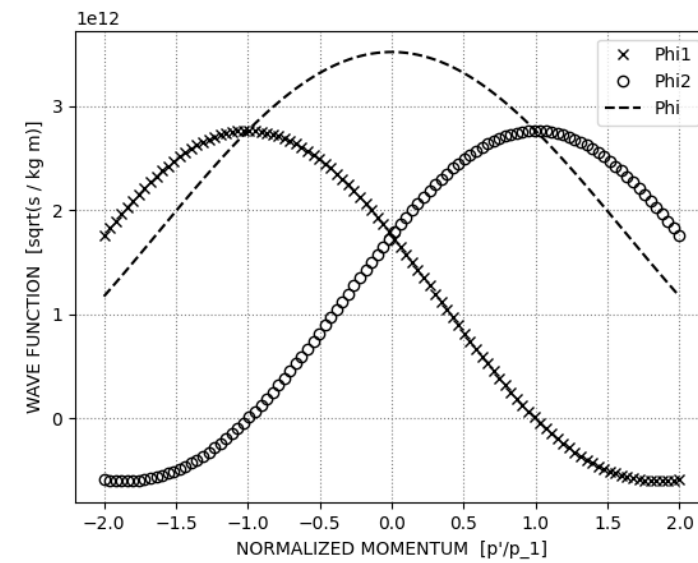
Sample Plots for Problem 1(d)

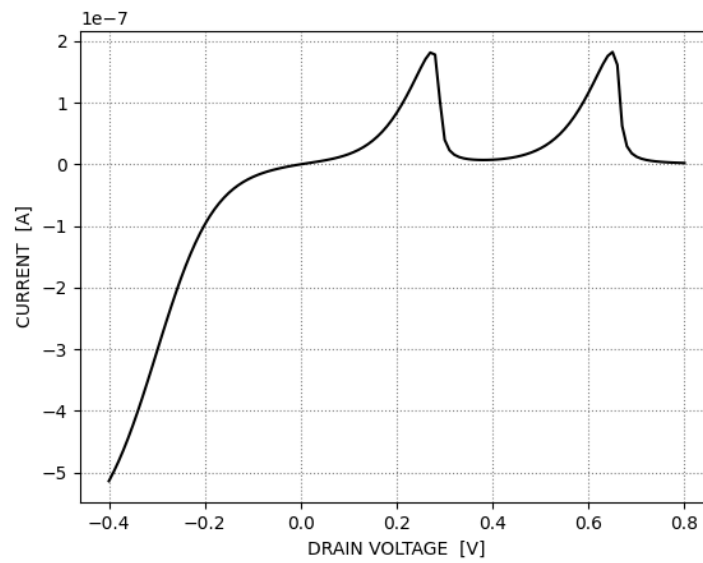


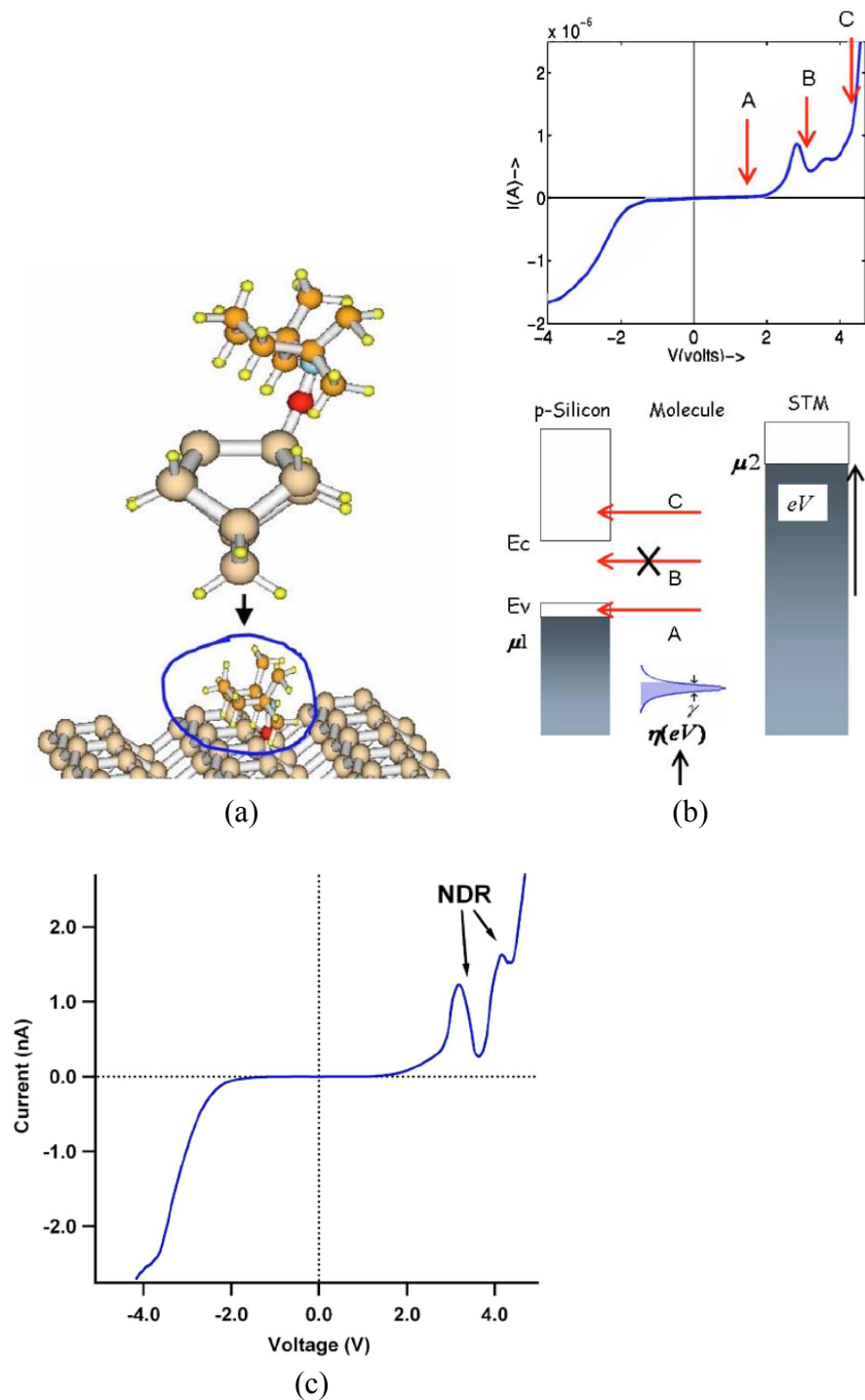
Sample Plots for Problem 2(c)



Sample Plots for Problem 3(b)(v)



Sample Plot for Problem 4(b)(i)



Published results from T. Rakshit, G. C. Liang, A. W. Ghosh, M. C. Hersam, and S. Datta, "Molecules on silicon: self-consistent first-principles theory and calibration to experiments," *Physical Review B (Condensed Matter and Materials Physics)*, pp. 125305-1-11, September 15, 2005.